

I'll always remember how satisfied I felt, watching my family play a game that I created and that worked as I intended it to.

That early experience had a lasting impact. There's real satisfaction in building something with a purpose, that solves a problem. The software I write now meets more significant needs than my childhood efforts did, but the sense of satisfaction I get from creating a program that works is still largely the same.

Who Is This Book For?

The goal of this book is to bring you up to speed with Python as quickly as possible so you can build programs that work—games, data visualizations, and web applications—while developing a foundation in programming that will serve you well for the rest of your life. *Python Crash Course* is written for people of any age who have never programmed in Python or have never programmed at all. This book is for those who want to learn the basics of programming quickly so they can focus on interesting projects, and those who like to test their understanding of new concepts by solving meaningful problems. *Python Crash Course* is also perfect for teachers at all levels who want to offer their students a project-based introduction to programming. If you're taking a college class and want a friendlier introduction to Python than the text you've been assigned, this book can make your class easier as well. If you're looking to change careers, *Python Crash Course* can help you make the transition into a more satisfying career track. It has worked well for a wide variety of readers, with a broad range of goals.

What Can You Expect to Learn?

The purpose of this book is to make you a good programmer in general and a good Python programmer in particular. You'll learn efficiently and adopt good habits as you gain a solid foundation in general programming concepts. After working your way through *Python Crash Course*, you should be ready to move on to more advanced Python techniques, and your next programming language will be even easier to grasp.

In Part I of this book, you'll learn basic programming concepts you need to know to write Python programs. These concepts are the same as those you'd learn when starting out in almost any programming language. You'll learn about different kinds of data and the ways you can store data in your programs. You'll build collections of data, such as lists and dictionaries, and you'll work through those collections in efficient ways. You'll learn to use `while` loops and `if` statements to test for certain conditions, so you can run specific sections of code while those conditions are true and run other sections when they're not—a technique that helps you automate many processes.

You'll learn to accept input from users to make your programs interactive, and to keep your programs running as long as the user wants. You'll explore how to write functions that make parts of your program reusable,